Yuchan Lee

Master's Student
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Research Interests

- Efficient System Design for Large-Scale Distributed Systems
- High-Performance Computing (HPC)
- Scheduling Heterogeneous Computing Resources

Education

- **Korea Aerospace University** *Sep 2024 - Aug 2026(Expected)*
  Department of Electronics and Information Engineering Goyang-si, Republic of Korea
  - Distributed & Big Data Computing Lab under the supervision of Prof. Jaehwan Lee
  - Master's student
  - Major: Computer Information Systems and Network
- **Korea Aerospace University** *Mar 2019 - Aug 2024*
  School of Electronics and Information Engineering Goyang-si, Republic of Korea
  - Bachelor of Engineering
  - Major: Computer Engineering

Publications

C=Conference, J=Journal

[C.1] **Yuchan Lee**, Sooho Jang, Sookwang Lee, Shinyoung Ahn, Jaehwan Lee (2025). **DEPUTY: A DPU-Based Network Offloading Architecture with Minimal CPU Involvement for Stable Network Performance**. In *2025 IEEE International Conference on Big Data (IEEE BigData 2025)*, Dec 2025, Macau SAR, China.

[C.2] Sooho Jang, Ahyeon Lim, **Yuchan Lee**, Sookwang Lee, Jaehwan Lee (2025). **P3P-Fed: Peer-to-Peer Personalized Federated Learning with DHT-based Local Clustering**. In *54th ACM International Conference on Parallel Processing (ICPP'25)*, Sep 2025, San Diego, USA.

[J.1] **Yuchan Lee**, Sookwang Lee, Jaehwan Lee (2024). **PyCAN: Open-source Python software of N-dimensional Content-Addressable Network**. *SoftwareX*, Vol. 28. DOI: 10.1016/j.softx.2024.101962 (SCIE)

Projects

- **Heterogeneous Accelerator Automatic Parallelization Technology Development** *Feb 2025 - Present*
  Tools: Python, C++, Pytorch, JAX, Huggingface, NPU Framework (TT-Metal, TT-NN)
  - Graduate Research Assistant (PI: Prof.Jaehwan Lee)
  - Developing an LLM scheduling technique that maximizes inference throughput in heterogeneous accelerator environments by factoring in the performance characteristics of GPU and NPU
  - Designing and implementing an algorithm to systematically explore and determine optimal distribution strategies for combined Tensor and Pipeline parallelism across the heterogeneous accelerators
- **Network Offloading System using Data Processing Units (DPU)** *Jul 2024 - Mar 2025*
  Tools: C, NVIDIA DPU Framework (DOCA)
  - Graduate Research Assistant (PI: Prof.Jaehwan Lee)
  - Mitigated CPU bottlenecks in network-request inference systems by developing a DPU-based network offloading architecture
  - Designed an optimized data path enabling the DPU to handle network processing and direct GPU memory DMA transfers, achieving up to 1.96x higher throughput in small data transfers compared to the existing DPU-based system
  - Research accepted for publication at the IEEE BigData 2025 conference

- **Development of a Scalable Python Content-Addressable Network** *Oct 2023 - Sep 2024*
  Tools: Python [ ]
  - Undergraduate Research Assistant (PI: Prof.Jaehwan Lee)
  - Developed PyCAN, the first open-source Python implementation of N-dimensional Content-Addressable Network (CAN) supporting a full set of DHT features to maintain P2P structure
  - Enhanced network stability via custom stabilization and verification algorithms, and validated scalability through simulation of 4,800 nodes
  - Research published in the SoftwareX journal (Vol. 28, 2024) (SCIE)
- **AI Convergence Capstone Design** *Sep 2023 - Jun 2024*
  Tools: Python
  - Advisors: Prof. Jaehwan Lee & Prof. Joonseon Ahn
  - Developed multi-dimensional coordinate and volume verification algorithms for validating the stability of a content-addressable network
- **Autonomous Vehicle Lane Recognition and Control System** *Feb 2023 - Nov 2023*
  Tools: Python, ROS, OpenCV
  - Team Leader
  - Developed a vision-based lane recognition and control system (OpenCV/Hybridnet/ROS) as a Developer in the KAUVOY autonomous vehicle club
- **Lockheed Martin Falcon Challenger** *Aug 2022 - Sep 2022*
  Tools: Python, OpenCV
  - Team Leader
  - Led the algorithm development for the Lockheed Martin Falcon Challenge (OpenCV-based drone mission), reaching the Semifinals

TEACHING EXPERIENCE

- **Graduate Teaching Assistant** *Sep 2024 - Present*
  Korea Aerospace University
  - Operating Systems
  - Software Convergence Design
  - Computer Information Networks
  - Big-data Computing and Applications
  - Advanced System Programming
  - Basic Circuits and Digital Laboratory
- **Undergraduate Teaching Assistant** *Mar 2024 - Jun 2024*
  Korea Aerospace University
  - Computational Thinking & Problem Solving

SKILLS

- **Programming Languages:** Python, C, C++
- **Database Systems:** SQL, Hadoop
- **Machine Learning:** Pytorch, JAX, Huggingface
- **Cloud Technologies:** Docker, VMWare
- **Specialized Area:** DOCA (NVIDIA DPU Framework), TT-Metal (Tenstorrent NPU low-level programming model), TT-NN (Tenstorrent NPU Neural Network Library), XLA (AI Graph Compiler), ROS (Robotics Operating System)

ADDITIONAL INFORMATION

Languages: Korean (Native Language), English (TOEFL: 96)